



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
FEBRUARY 2022**

Course : EC3105 (C Programming)

Time: 8.00am – 11.00 am
(3 hours)

Lecturer : Ts. Harlina Harun

Date : 13 June 2022

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 3 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. The following questions refer to the screen illustrated below that shows the partial result of ls-al command. State appropriate UNIX command(s) to answer the following questions.

```
-rwxr-xr-x 1 alexa student 23068 Jan 26 19:98 archive.sh
-rw-rw-r-- 1 alexa student 12878 Feb 24 21:58 personal.txt
-rw-rw-r-- 1 alexa student 2645 Jun 30 08:48 personnel.txt
-rw-r--r-- 1 alexa student 168 Apr 17 11:51 university.html
drwxr-xr-x 2 alexa student 1024 Mar 18 16:27 assign
-rwxr-xr-x 1 alexa student 2645 Aug 4 11:03 program
```

Quotes (" ") were added to highlight names and can be excluded.

- i. Search for the expression "grade" in "university.html". (3 MARKS)
 - ii. Give full permission to the "personnel.txt" (3 MARKS)
 - iii. Remove all permissions of the "personal.txt" (3 MARKS)
 - iv. Run "program" executable file and take input from "customer" file. (3 MARKS)
- b. Write a C program that asks the user to enter a file name and delete a filename using Linux command according to the input file name. (13 MARKS)
2. a. Define the following terms:
- i. Strings (2 MARKS)
 - ii. Call by value (2 MARKS)
 - iii. Variables (2 MARKS)
 - iv. Datatypes (2 MARKS)
 - v. Function definition (2 MARKS)
- b. Write a C program to find how many consonants and vowels in the string below. (15 MARKS)

```
char name []="GOOD MORNING";
```

3. a. What is the output of the following C Programs?

i. Suppose the input values are 10, 10, 20, 30, 30.

(5 MARKS)

```
#include<stdio.h>

int main(void)
{
    int arr[5], i;
    printf ("\n Enter five values: ");
    for (i=0; i<5; i++)
        scanf ("%d", &arr[i]);
    printf ("\n The output is:");
    for (i=4; i>=0; i--)
        printf ("%4d", *(arr+i));
}
```

ii. Suppose the input is 4.

(5 MARKS)

```
#include<stdio.h>

void cbr(int *np);

int main(void)
{
    int num;
    printf("Please Enter a number\t");
    scanf("%d", &num);
    printf("The original value of number is
%d", num++);
    cbr(&num);
    printf("\nThe new value of number
is:\t%d\n", ++num);
    return 0;
}

void cbr(int *np)
{
    printf("\nThe value in function is:
\t%d", *np);
    *np = *np * (*np) * (*np);
}
```

b. Explain **THREE (3)** types of programming errors

(6 MARKS)

- c. Write a program using nested *for loop* to produce the following output:

(9 MARKS)

10	9	8	7	6
9	8	7	6	5
8	7	6	5	4
7	6	5	4	3
6	5	4	3	2

4. Write a C program that reads some integer numbers from the input and saves it in an array. Then, the program calls a function that finds the minimum number value and returns it to the main program. Finally, the program prints the minimum number value. Assume that the maximum number of integer numbers is 10.

(25 MARKS)

Example of output:

```
Type list[0]: 20
Type list[1]: 10
Type list[2]: 4
Type list[3]: 30
Type list[4]: 21

Minimum is: 4
```

-END OF QUESTION PAPER-